Mathematical Properties of the Affine-DDPNM

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All results are stated at the finite-dimensional discrete level. Local-kernel and global-SPD conclusions are conditional on explicit rank and anchoring assumptions. Galerkin optimality is proved relative to a separately defined broken-pressure full-traction-trace system and only in its energy seminorm. Equivalence with the monolithic continuous-pressure Taylor–Hood method is *not* asserted.

1 Discrete setting and sign convention

1.1 Local Stokes discretization

For pore Ω_i, let $A_i \in \mathbb{R}^{n_{u,i}\times n_{u,i}}$ and $B_i \in \mathbb{R}^{n_{p,i}\times n_{u,i}}$ correspond to

$$a_i(\boldsymbol{u},\boldsymbol{v}) = \nu\int_{\Omega_i} \nabla\boldsymbol{u} : \nabla\boldsymbol{v}\,\mathrm{d}x, \qquad b_i(\boldsymbol{v},q) = -\int_{\Omega_i} q\,\nabla\cdot\boldsymbol{v}\,\mathrm{d}x.$$

The local saddle-point system reads

$$K_i \begin{bmatrix} U \\ P \end{bmatrix} := \begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} U \\ P \end{bmatrix} = \begin{bmatrix} f \\ \mathbf{0} \end{bmatrix}. \tag{1}$$

No pressure stabilisation is used.

Assumption 1.1 (Local algebraic stability)**.** *For every pore i:*
(a) A_i is symmetric positive definite;
(b) B_i has full row rank (discrete inf–sup; not implied by (a));
(c) the coefficient vector $\mathbf{1}_i^p$ of the constant pressure $q_h \equiv 1$ belongs to the local pressure space.

For the standard conforming velocity space, condition (a) follows, for example, if Ω_i is connected and Lipschitz and its essential no-slip boundary portion $\Gamma_{w,i} \subset \partial\Omega_i$ has positive surface measure. The Poincaré–Friedrichs inequality then excludes nonzero constant-velocity null modes.

1.2 Interface frames and oriented directions

Each internal interface γ_k is planar and is assigned a global orthonormal frame $\{\boldsymbol{n}_k, \boldsymbol{t}_{k,1}, \boldsymbol{t}_{k,2}\}$ with $\boldsymbol{n}_k$ directed from the first pore of the pair to the second. Define $\sigma_{i,k} = +1$ if Ω_i is the first member and -1 otherwise. The **oriented inward-traction directions** are

$$\boldsymbol{d}_{i,k,n} = \boldsymbol{n}_{i,k} = \sigma_{i,k}\boldsymbol{n}_k, \qquad \boldsymbol{d}_{i,k,t_\nu} = -\sigma_{i,k}\,\boldsymbol{t}_{k,\nu} \ \ (\nu = 1,2). \tag{2}$$

The minus sign in the tangential definition is a pure convention chosen so that the assembled load direction $-\boldsymbol{d}_{i,k,t_\nu} = +\sigma_{i,k}\boldsymbol{t}_{k,\nu}$ matches the code's `side_sign * tangent`. Crucially,

$$\boldsymbol{d}_{j,k,\beta} = -\boldsymbol{d}_{i,k,\beta} \qquad (\beta = n, t_1, t_2). \tag{3}$$

Half-span–normalised in-plane coordinates: $s_k(\boldsymbol{x}) = ((\boldsymbol{x}-\boldsymbol{c}_k)\cdot\boldsymbol{t}_{k,1})/a_k$, $t_k(\boldsymbol{x}) = ((\boldsymbol{x}-\boldsymbol{c}_k)\cdot\boldsymbol{t}_{k,2})/b_k$, with $\varphi_0 = 1$, $\varphi_s = s_k$, $\varphi_t = t_k$, $\mathcal{M} = \{0,s,t\}\times\{n,t_1,t_2\}$.

1.3 Traction Ansatz and primitive loads

The unknown coefficients parameterize the **inward** traction:

$$-\boldsymbol{\sigma}(\boldsymbol{u}_i, p_i)\, \boldsymbol{n}_{i,k} = \sum_{(\alpha,\beta)\in\mathcal{M}} \lambda_{k,\alpha,\beta}\, \varphi_\alpha\, \boldsymbol{d}_{i,k,\beta}. \tag{4}$$

The physical outward traction in the weak form is therefore $\boldsymbol{\sigma}\boldsymbol{n}_{i,k} = -\sum \lambda\varphi\boldsymbol{d}_{i,k,\beta}$. The primitive load column for mode (α,β) on γ_k is

$$\big[\boldsymbol{f}_{i,k}^{(\alpha,\beta)}\big]_j = -\int_{\gamma_k} \varphi_\alpha\, \boldsymbol{d}_{i,k,\beta}\cdot\boldsymbol{\phi}_j^{(i)}\,\mathrm{d}S. \tag{5}$$

The leading minus sign is essential: it converts the inward-traction coefficient into the physical outward-traction boundary integral $\int_{\partial\Omega_i}(\boldsymbol{\sigma}\boldsymbol{n})\cdot\boldsymbol{v}\,\mathrm{d}S$.

Let $L_i \in \mathbb{R}^{n_{u,i}\times n_i}$ collect all primitive columns, $n_i = 9m_i^u + m_i^k$, including one constant-normal column per known boundary port with prescribed coefficient $\lambda_{i,\gamma}^k = p_b$.

1.4 Compliance map and weighted velocity moments

Define

$$M_i = B_i A_i^{-1} B_i^\top, \qquad H_i = A_i^{-1} - A_i^{-1}B_i^\top M_i^{-1} B_i A_i^{-1}. \tag{6}$$

Under Assumption 1.1, M_i is SPD.

Lemma 1.2 (Properties of H_i)**.** *H_i is symmetric positive semidefinite and satisfies $H_i^\top A_i H_i = H_i$, $\mathrm{Ker}(H_i) = \mathrm{Range}(B_i^\top)$. For any f, the velocity component of the solution to (1) is $U = H_i f$ and $B_i U = 0$.*

Proof. Let $C_i = A_i^{-1/2}B_i^\top$ and $\Pi_i = C_i(C_i^\top C_i)^{-1}C_i^\top$. Then $M_i = B_iA_i^{-1}B_i^\top = C_i^\top C_i$ is SPD (Assumption 1.1(b) and $A_i \succ 0$). A direct computation gives

$$H_i = A_i^{-1/2}(I-\Pi_i)A_i^{-1/2}.$$

Since $I-\Pi_i$ is an orthogonal projector (symmetric idempotent), H_i is symmetric positive semidefinite, with $\mathrm{Ker}(H_i) = A_i^{1/2}\,\mathrm{Range}(\Pi_i) = \mathrm{Range}(B_i^\top)$ and $H_i^\top A_i H_i = A_i^{-1/2}(I-\Pi_i)A_i^{-1/2} = H_i$. Solving (1) by block elimination yields $U = H_i f$ and $P = M_i^{-1}B_iA_i^{-1}f$; the second block row then gives $B_iU = B_iH_if = 0$. □

The **zero-stabilisation assumption** $\varepsilon = 0$ is required for Lemma 1.2 ($B_iH_i = 0$, discrete incompressibility) and for the exact mass conservation statements below. The Affine-DDPNM local-response computations and the associated exact mass-conservation results use $\varepsilon = 0$. The current Exact FE-trace comparator uses a separate stabilised matrix with $\varepsilon = 10^{-10}$ and is not covered by Lemma 1.2 or the exact mass-conservation theorem. The case $\varepsilon > 0$ requires a modified analysis (the second block row becomes $B_iU_i - \varepsilon M_{p,i}P_i = 0$, breaking exact $B_iU_i = 0$).

The local compliance matrix is

$$\mathbf{G}_i = L_i^\top H_i L_i \;\in\; \mathbb{R}^{n_i\times n_i}, \tag{7}$$

symmetric positive semidefinite. Partition: $\mathbf{G}_i = \left[\begin{smallmatrix}\mathbf{G}_i^{uu} & \mathbf{G}_i^{uk}\\ \mathbf{G}_i^{ku} & \mathbf{G}_i^{kk}\end{smallmatrix}\right]$.

The weighted velocity moment for mode (α,β) on interface γ_k of Ω_i is

$$W_{i,k}^{(\alpha,\beta)}(U_i) = \int_{\gamma_k} \varphi_\alpha\, \boldsymbol{u}_{i,h}\cdot\boldsymbol{d}_{i,k,\beta}\,\mathrm{d}S. \tag{8}$$

From the load convention (5), $[\mathbf{G}_i\boldsymbol{\lambda}_i]_{k,\alpha,\beta} = -W_{i,k}^{(\alpha,\beta)}(U_i)$, so $\mathbf{G}_i$ is the *negative* moment response.

The global Schur system, enforcing $W_{i,k}^{(\alpha,\beta)} + W_{j,k}^{(\alpha,\beta)} = 0$ for all nine modes:

$$\mathbf{S}\,\boldsymbol{\lambda} = \boldsymbol{F}, \qquad \mathbf{S} = \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u, \qquad \boldsymbol{F} = -\sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \boldsymbol{\lambda}^k. \tag{9}$$

2 Local kernel — conditional characterization

Lemma 2.1 (Symmetry and energy identity)**.** $\boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i = U_i^\top A_i U_i \geq 0$, *where* $U_i = H_i L_i \boldsymbol{\lambda}_i$.

Proof. $U_i = H_i L_i \boldsymbol{\lambda}_i$ by Lemma 1.2. Then $\boldsymbol{\lambda}_i^\top L_i^\top H_i L_i \boldsymbol{\lambda}_i = (H_i L_i \boldsymbol{\lambda}_i)^\top A_i (H_i L_i \boldsymbol{\lambda}_i)$ using $H_i^\top A_i H_i = H_i$. $\square$

Lemma 2.2 (Exact kernel formula)**.** $\mathrm{Ker}(\mathbf{G}_i) = \mathrm{Ker}(H_i L_i) = \{\boldsymbol{\lambda}_i : L_i \boldsymbol{\lambda}_i \in \mathrm{Range}(B_i^\top)\}$.

Proof. From Lemma 2.1, $\mathbf{G}_i \boldsymbol{\lambda}_i = \mathbf{0}$ iff $H_i L_i \boldsymbol{\lambda}_i = \mathbf{0}$. By Lemma 1.2, $H_i f = 0$ iff $f \in \mathrm{Range}(B_i^\top)$. Hence $H_i L_i \boldsymbol{\lambda}_i = 0$ iff $L_i \boldsymbol{\lambda}_i \in \mathrm{Range}(B_i^\top)$. Note: no claim that H_i is SPD on $\mathrm{Range}(L_i)$ is made or needed; that claim would be false because $\mathrm{Range}(L_i) \ni B_i^\top \mathbf{1}_i^p \in \mathrm{Ker}(H_i)$. $\square$

Let $\mathbf{1}_{i,0n} \in \mathbb{R}^{n_i}$ have entry 1 at every constant-normal mode $(\gamma, 0, n)$ (internal and boundary) and 0 elsewhere.

Lemma 2.3 (Constant-pressure null vector)**.** $L_i \mathbf{1}_{i,0n} = B_i^\top \mathbf{1}_i^p$. *Hence* $\mathbf{1}_{i,0n} \in \mathrm{Ker}(\mathbf{G}_i)$.

Proof. Summing the constant-normal load columns over all ports (internal $\mathcal{U}_i$ and boundary $\mathcal{K}_i$):

$$
\begin{aligned}
[L_i \mathbf{1}_{i,0n}]_j &= -\sum_{\gamma \in \mathcal{U}_i \cup \mathcal{K}_i} \int_\gamma \boldsymbol{\phi}_j^{(i)} \cdot \boldsymbol{n}_i \, \mathrm{d}S \\
&= -\int_{\partial \Omega_i} \boldsymbol{\phi}_j^{(i)} \cdot \boldsymbol{n}_i \, \mathrm{d}S + \int_{\Gamma_{i,\mathrm{wall}}} \boldsymbol{\phi}_j^{(i)} \cdot \boldsymbol{n}_i \, \mathrm{d}S \\
&= -\int_{\Omega_i} \nabla \cdot \boldsymbol{\phi}_j^{(i)} \, \mathrm{d}x = [B_i^\top \mathbf{1}_i^p]_j,
\end{aligned}
$$

where the wall integral vanishes because each velocity basis function $\boldsymbol{\phi}_j^{(i)}$ is zero on the no-slip wall $\Gamma_{i,\mathrm{wall}}$ (not because the physical traction vanishes there). Then Lemma 2.2 applies. $\square$

Assumption 2.4 (No non-constant pressure-load dependency)**.** *For each pore,*

$$
L_i \boldsymbol{\lambda}_i \in \mathrm{Range}(B_i^\top) \implies \boldsymbol{\lambda}_i = c\, \mathbf{1}_{i,0n} \ \text{ for some } c \in \mathbb{R}. \tag{10}
$$

Equivalently, $\mathrm{rank}(H_i L_i) = n_i - 1$.

Theorem 2.5 (Conditional local-kernel characterization)**.** *Under Assumptions 1.1 and 2.4,* $\mathrm{Ker}(\mathbf{G}_i) = \mathrm{span}\{\mathbf{1}_{i,0n}\}$.

Proof. Lemma 2.3 gives $\supseteq$. For $\subseteq$, any $\boldsymbol{\lambda}_i \in \mathrm{Ker}(\mathbf{G}_i)$ satisfies $L_i \boldsymbol{\lambda}_i \in \mathrm{Range}(B_i^\top)$ by Lemma 2.2. Assumption 2.4 forces $\boldsymbol{\lambda}_i = c\, \mathbf{1}_{i,0n}$. $\square$

Remark 2.6 (Interpretation of the rank condition)**.** *Equation (10) states that the n_i responses $H_i \boldsymbol{f}_i^{(\mu)}$ satisfy exactly one linear dependency—the constant-pressure relation of Lemma 2.3. It does* not *assert that these responses have codimension 1 in* $\mathrm{Range}(H_i)$*; that codimension is* $\mathrm{rank}(H_i) - (n_i - 1)$ *and is generally large. Per-face linear independence of the nine scalar/vector functions is necessary but not sufficient for Assumption 2.4: it does not exclude cross-interface dependencies or dependencies introduced by the incompressible projection H_i.*

3 Global Schur complement — SPD and mass conservation

For pore i, let $J_i^u : \mathbb{R}^{9 m_i^u} \to \mathbb{R}^{n_i}$ be the zero extension into the known boundary slots: $J_i^u(z) = (z, \mathbf{0})$. The global energy is

$$
\boldsymbol{\lambda}^\top \mathbf{S} \boldsymbol{\lambda} = \sum_{i=1}^{N} (J_i^u R_i^u \boldsymbol{\lambda})^\top \mathbf{G}_i (J_i^u R_i^u \boldsymbol{\lambda}). \tag{11}
$$

Assumption 3.1 (Graph anchoring)**.** *Every connected component of the pore adjacency graph contains at least one pore with at least one prescribed-traction (inlet/outlet) port.*

Theorem 3.2 (Global SPD and unique solvability)**.** *Under Assumptions 1.1, 2.4, and 3.1,* $\mathbf{S}$ *is SPD.*

Proof. Symmetry and positive semidefiniteness follow from (11) and Lemma 2.1. Suppose $\boldsymbol{\lambda}^\top \mathbf{S} \boldsymbol{\lambda} = 0$. Every summand vanishes, so Theorem 2.5 gives $J_i^u R_i^u \boldsymbol{\lambda} = c_i \mathbf{1}_{i,0n}$ for each pore.

If pore i has a boundary port, its slot in $J_i^u R_i^u \boldsymbol{\lambda}$ is zero (by the zero extension) but the corresponding slot in $\mathbf{1}_{i,0n}$ is 1; hence $c_i = 0$.

For each internal interface γ_k shared by pores i and j, let $E_{i,k} : \mathbb{R}^{9m} \to \mathbb{R}^9$ extract the coefficient block $\boldsymbol{\lambda}_k \in \mathbb{R}^9$ from the global vector $\boldsymbol{\lambda}$. By construction of the Boolean restriction maps, $E_{i,k} = E_{j,k}$: both pores read the *same* nine global coefficients on γ_k. (The opposite sign of the physical traction is encoded in $\boldsymbol{d}_{j,k,\beta} = -\boldsymbol{d}_{i,k,\beta}$, not in the coefficients.) The constant-normal components thus satisfy $c_i = [E_{i,k}\, \boldsymbol{\lambda}]_{0,n} = [E_{j,k}\, \boldsymbol{\lambda}]_{0,n} = c_j$. Hence $c_i = c_j$ for every adjacent pair; the constants are uniform on each connected graph component. Assumption 3.1 ensures every component contains a boundary pore where $c = 0$; hence all $c_i = 0$ and $\boldsymbol{\lambda} = \mathbf{0}$. ☐

Remark 3.3 (Floating basins are distinct from anchoring)**.** ***Local coercivity*** *(Assumption 1.1(a)) fails for pores without a solid-wall facet (*floating basins*); these are merged into neighbours before assembly.* ***Graph anchoring*** *(Assumption 3.1) is a separate condition requiring an inlet/outlet port in each connected component. Merging a floating basin restores coercivity but does* not *supply graph anchoring.*

Theorem 3.4 (Discrete mass conservation)**.** *Under Assumption 1.1, each local reconstructed velocity satisfies* $\sum_k W_{i,k}^{(0,n)} = 0$*. If, in addition, the global Schur system* (9) *admits a solution* $\boldsymbol{\lambda}$*, then the assembled velocity field satisfies* $\int_{\partial\Omega} \boldsymbol{u}_h^{DD} \cdot \boldsymbol{n}\, \mathrm{d}S = 0$.

Proof. $B_i U_i = 0$ and the constant test function $\mathbf{1}_i^p$ give the local balance. The Schur equations enforce $W_{i,k}^{(0,n)} + W_{j,k}^{(0,n)} = 0$; summing over pores cancels internal fluxes, leaving only the boundary contributions which sum to zero. ☐

A Error estimate as a Galerkin projection

We define a broken-pressure full-traction-trace (C-DD) reference system and prove that the Affine-DDPNM is its Galerkin restriction. Unless stated otherwise, Assumption 1.1 (local algebraic stability: $A_i \succ 0$, B_i full row rank, constant pressure representable) remains in force throughout this appendix. We also assume that the full C-DD Schur system is consistent, i.e. $\widehat{\boldsymbol{F}} \in \mathrm{Range}(\widehat{\mathbf{S}})$. Its derivation from a fully specified broken-pressure mixed Stokes formulation is outside the scope of this appendix.

Scope. All results in this appendix assume **planar** interfaces with **matching** trace spaces (Assumption A.1). They do *not* apply to non-planar interfaces (e.g. watershed partitions with bent, faceted interfaces or per-facet normals $\boldsymbol{n}(x)$). Numerical results on watershed geometries are not covered by the optimality guarantee of Theorem A.3.

This reference system uses *discontinuous* pressure across pores ($Q_h^{\mathrm{br}} = \bigoplus_i Q_{i,h}$); it is not the monolithic continuous-P_1 Taylor–Hood system.

A.1 Full-traction-trace (C-DD) system

On interface γ_k, let $\{\psi_{k,\ell}\}_{\ell=1}^{m_k}$ be a scalar trace basis (the restriction of the $[P_2]^3$ velocity trace). The C-DD variable $\boldsymbol{\xi}$ is defined to represent the **inward** traction $-\boldsymbol{\sigma n}$ at mesh nodes, matching the affine convention (4). Define coupling blocks using the same oriented directions as in (2):

$$[N_{i,k}^{\beta}]_{\ell,j} = \int_{\gamma_k} \psi_{k,\ell}\, \boldsymbol{d}_{i,k,\beta} \cdot \boldsymbol{\phi}_j^{(i)}\, \mathrm{d}S, \qquad \beta \in \{n, t_1, t_2\}. \tag{12}$$

For an inward-traction nodal vector $\boldsymbol{\xi}_{i,k} = (\xi^n; \xi^{t_1}; \xi^{t_2}) \in \mathbb{R}^{3m_k}$, the physical outward traction is $\boldsymbol{\sigma n} = -\boldsymbol{\xi}$; the weak-form boundary load is therefore $-N_i^{\top}\boldsymbol{\xi}_i$, matching the load convention (5).

The local C-DD Schur complement is $\widehat{\mathbf{S}}_i = N_i H_i N_i^{\top}$. Boolean assembly uses standard maps $\widehat{R}_i^u$ (no additional sign factor needed: the orientation $\boldsymbol{d}_{j,k,\beta} = -\boldsymbol{d}_{i,k,\beta}$ from (3) is already encoded in the coupling blocks $N_{i,k}^{\beta}$ and the globally shared coefficient blocks). Assembly yields

$$\widehat{\mathbf{S}}\,\boldsymbol{\xi}^{\mathrm{full}} = \widehat{\boldsymbol{F}}, \qquad \widehat{\mathbf{S}} = \sum_i (\widehat{R}_i^u)^{\top} \widehat{\mathbf{S}}_i^{uu} \widehat{R}_i^u. \tag{13}$$

$\widehat{\mathbf{S}}$ is symmetric positive semidefinite. Define $\|x\|_{\widehat{\mathbf{S}}} = (x^{\top}\widehat{\mathbf{S}}x)^{1/2}$; without a full-trace kernel theorem this is a seminorm.

Let $m_{\Gamma,i}^k := 3\sum_{\gamma\in\mathcal{K}_i} m_{i,\gamma}$ be the total number of C-DD boundary trace DOFs on Ω_i, where $m_{i,\gamma}$ is the number of scalar trace nodes on boundary port γ. The C-DD boundary data vector is $\boldsymbol{\xi}_i^k \in \mathbb{R}^{m_{\Gamma,i}^k}$, collecting the known inward-traction nodal values on all boundary ports of Ω_i. The corresponding coupling blocks $N_i^k \in \mathbb{R}^{m_{\Gamma,i}^k \times n_{u,i}}$ are assembled from (12) restricted to the boundary ports, and the C-DD right-hand side is

$$\widehat{\boldsymbol{F}} = -\sum_{i=1}^{N} (\widehat{R}_i^u)^{\top} N_i^u H_i (N_i^k)^{\top} \boldsymbol{\xi}_i^k. \tag{14}$$

In the affine system, each boundary port carries only the single constant-normal mode. We define the boundary restriction map $P_i^k \in \mathbb{R}^{m_{\Gamma,i}^k \times m_i^k}$ columnwise: each boundary port $\gamma \in \mathcal{K}_i$ contributes one column, the L^2-projection of $\varphi_0 = 1$ onto the scalar trace basis on that port, placed in the normal block-row. The boundary nodal inward traction is then $\boldsymbol{\xi}_i^k = P_i^k R_i^k \boldsymbol{\lambda}^k$, and the load identity extends to $L_i^k = -(N_i^k)^{\top} P_i^k$.

Assumption A.1 (Matching interface trace spaces)**.** *For every internal interface γ_k shared by pores i and j, the two local velocity trace spaces coincide: $\mathbf{V}_{i,h}|_{\gamma_k} = \mathbf{V}_{j,h}|_{\gamma_k}$, with a shared scalar trace basis $\{\psi_{k,\ell}\}$ and a one-to-one DOF correspondence between the two sides. The Boolean assembly maps $\widehat{R}_i^u$ operate on this shared nodal basis.*

A.2 Affine restriction and algebraic equivalence

Let M_k be the face mass matrix, $[M_k]_{\ell,\ell'} = \int_{\gamma_k} \psi_{k,\ell}\psi_{k,\ell'}\, \mathrm{d}S$. Since $\{\psi_{k,\ell}\}_{\ell=1}^{m_k}$ is a linearly independent $L^2(\gamma_k)$ basis, M_k is symmetric positive definite. Define $p_k^{\alpha} = M_k^{-1} b_k^{\alpha}$, $(b_k^{\alpha})_{\ell} = \int_{\gamma_k} \varphi_{\alpha}\psi_{k,\ell}\, \mathrm{d}S$, $\alpha \in \{0, s, t\}$. For the unconstrained full $[P_2]^3$ trace space on a triangular face, each φ_{α} belongs to the scalar trace space. Hence its L^2-projection is exact: $p_{k,h}^{\alpha} = \varphi_{\alpha}$ pointwise on γ_k. If trace DOFs are eliminated by no-slip wall constraints, only the mass-matrix orthogonality $\int_{\gamma_k} (\varphi_{\alpha} - p_{k,h}^{\alpha})\, q_h\, \mathrm{d}S = 0$ for all $q_h \in T_{k,h}$ is guaranteed; the proof below uses only this orthogonality and does not rely on pointwise equality.

With coefficient order $(0,n), (0,t_1), (0,t_2), (s,n), (s,t_1), (s,t_2), (t,n), (t,t_1), (t,t_2)$, the correct restriction matrix is

$$P_k^{\mathrm{aff}} = \begin{bmatrix} p_k^0 & 0 & 0 & p_k^s & 0 & 0 & p_k^t & 0 & 0 \\ 0 & p_k^0 & 0 & 0 & p_k^s & 0 & 0 & p_k^t & 0 \\ 0 & 0 & p_k^0 & 0 & 0 & p_k^s & 0 & 0 & p_k^t \end{bmatrix} \in \mathbb{R}^{3m_k \times 9}. \tag{15}$$

Each column occupies only the block-row matching its direction β. The global restriction is $\mathcal{P}_{\text{aff}} = \bigoplus_k P_k^{\text{aff}} : \mathbb{R}^{9m} \to \mathbb{R}^{m_\Gamma}$, where $m_\Gamma = 3\sum_{k=1}^{m} m_k$ is the total number of C-DD interface traction DOFs (three components per trace node per interface).

Lemma A.2 (Algebraic equivalence). *Under Assumption A.1, $\mathbf{S} = \mathcal{P}_{\mathit{aff}}^\top \widehat{\mathbf{S}}\, \mathcal{P}_{\mathit{aff}}$, $\boldsymbol{F} = \mathcal{P}_{\mathit{aff}}^\top \widehat{\boldsymbol{F}}$.*

Proof. **Step 1: load–coupling identity.** Let $p_{k,h}^\alpha := \sum_{\ell=1}^{m_k} (p_k^\alpha)_\ell\, \psi_{k,\ell} \in \operatorname{span}\{\psi_{k,\ell}\}$ be the L^2-projection of φ_α onto the scalar trace space $T_{k,h} := \operatorname{span}\{\psi_{k,\ell}\}_{\ell=1}^{m_k}$. By construction, $\int_{\gamma_k} (\varphi_\alpha - p_{k,h}^\alpha)\, q_h \,\mathrm{d}S = 0$ for every $q_h \in T_{k,h}$. For a planar interface γ_k, the direction vectors $\boldsymbol{d}_{i,k,\beta}$ are constant, and the restriction of each velocity basis function satisfies $\boldsymbol{d}_{i,k,\beta} \cdot \boldsymbol{\phi}_j^{(i)}|_{\gamma_k} \in T_{k,h}$. Hence the L^2-projection can be moved from the scalar shape to the full integrand:

$$\int_{\gamma_k} \varphi_\alpha\, \boldsymbol{d}_{i,k,\beta} \cdot \boldsymbol{\phi}_j^{(i)} \,\mathrm{d}S = \int_{\gamma_k} p_{k,h}^\alpha\, \boldsymbol{d}_{i,k,\beta} \cdot \boldsymbol{\phi}_j^{(i)} \,\mathrm{d}S.$$

Expanding $p_{k,h}^\alpha$ in the trace basis and comparing with (5) and (12) yields the column-wise identity $L_i^u = -(N_i^u)^\top P_i^{\text{aff}}$, where the minus sign comes from the load convention (5). The boundary counterpart $L_i^k = -(N_i^k)^\top P_i^k$ follows analogously.

Step 2: quadratic cancellation. From Step 1, $\mathbf{G}_i^{uu} = (L_i^u)^\top H_i L_i^u = (P_i^{\text{aff}})^\top N_i^u H_i (N_i^u)^\top P_i^{\text{aff}}$ (the minus signs cancel quadratically). Summing with the Boolean maps and using the commutativity of the block-diagonal restriction with the global interface indexing gives $\mathbf{S} = \mathcal{P}_{\text{aff}}^\top \widehat{\mathbf{S}}\, \mathcal{P}_{\text{aff}}$. For the forcing term: the boundary counterpart of the load identity is $L_i^k = -(N_i^k)^\top P_i^k$. From (14) and $\boldsymbol{\xi}_i^k = P_i^k R_i^k \boldsymbol{\lambda}^k$,

$$\begin{aligned}
\mathcal{P}_{\text{aff}}^\top \widehat{\boldsymbol{F}} &= \sum_i (R_i^u)^\top (P_i^{\text{aff}})^\top \big(-N_i^u H_i (N_i^k)^\top P_i^k R_i^k \boldsymbol{\lambda}^k\big) \\
&= -\sum_i (R_i^u)^\top \big(-(L_i^u)^\top\big) H_i \big(-L_i^k\big) R_i^k \boldsymbol{\lambda}^k \\
&= -\sum_i (R_i^u)^\top (L_i^u)^\top H_i L_i^k R_i^k \boldsymbol{\lambda}^k = -\sum_i (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \boldsymbol{\lambda}^k = \boldsymbol{F},
\end{aligned}$$

where we used $L_i^u = -(N_i^u)^\top P_i^{\text{aff}}$, $L_i^k = -(N_i^k)^\top P_i^k$, and the definition $\mathbf{G}_i^{uk} = (L_i^u)^\top H_i L_i^k$. The three minus signs from the load identities and the C-DD right-hand side produce a net single minus sign, exactly matching the definition of $\boldsymbol{F}$ in (9). □

A.3 Error estimate and enrichment

Consistency of the restricted system. Full consistency $\widehat{\mathbf{S}}\, \boldsymbol{\xi}^{\text{full}} = \widehat{\boldsymbol{F}}$ implies that the restricted system $\mathbf{S}\, \boldsymbol{\lambda} = \boldsymbol{F}$ is also consistent. Indeed, let $A = \widehat{\mathbf{S}}^{1/2} \mathcal{P}_{\text{aff}}$; then $\mathbf{S} = A^\top A$ and $\operatorname{Range}(\mathbf{S}) = \operatorname{Range}(A^\top)$. From full consistency, $\boldsymbol{F} = \mathcal{P}_{\text{aff}}^\top \widehat{\boldsymbol{F}} = \mathcal{P}_{\text{aff}}^\top \widehat{\mathbf{S}}\, \boldsymbol{\xi}^{\text{full}} = A^\top \widehat{\mathbf{S}}^{1/2} \boldsymbol{\xi}^{\text{full}} \in \operatorname{Range}(A^\top) = \operatorname{Range}(\mathbf{S})$. Hence there exists $\boldsymbol{\lambda}^{\text{aff}}$ solving $\mathbf{S}\, \boldsymbol{\lambda}^{\text{aff}} = \boldsymbol{F}$.

Non-uniqueness. If $\widehat{\mathbf{S}}$ is only semidefinite, full solutions differ by elements of $\ker(\widehat{\mathbf{S}})$, and affine coefficient solutions differ by elements of $\ker(\mathbf{S})$. The vector $\boldsymbol{\xi}^{\text{aff}} = \mathcal{P}_{\text{aff}}\, \boldsymbol{\lambda}^{\text{aff}}$ is unique only modulo $\ker(\widehat{\mathbf{S}})$: any two solutions $\boldsymbol{\lambda}, \boldsymbol{\lambda}'$ satisfy $\mathcal{P}_{\text{aff}}(\boldsymbol{\lambda} - \boldsymbol{\lambda}') \in \ker(\widehat{\mathbf{S}})$, so $\|\mathcal{P}_{\text{aff}} \boldsymbol{\lambda} - \mathcal{P}_{\text{aff}} \boldsymbol{\lambda}'\|_{\widehat{\mathbf{S}}} = 0$. The equivalence class $[\boldsymbol{\xi}^{\text{aff}}] \in \mathbb{R}^{m_\Gamma} / \ker(\widehat{\mathbf{S}})$ is uniquely determined, although the vector representative need not be. The seminorm values in the optimality statement below are therefore independent of the choice of representative solution.

Theorem A.3 (Best approximation in the full-trace energy seminorm). *Under Assumption A.1, let $\boldsymbol{\xi}^{\mathit{full}}$ be any solution to $\widehat{\mathbf{S}}\, \boldsymbol{\xi}^{\mathit{full}} = \widehat{\boldsymbol{F}}$. Let $\boldsymbol{\lambda}^{\mathit{aff}}$ be any solution to $\mathbf{S}\, \boldsymbol{\lambda}^{\mathit{aff}} = \boldsymbol{F}$ and set $\boldsymbol{\xi}^{\mathit{aff}} = \mathcal{P}_{\mathit{aff}}\, \boldsymbol{\lambda}^{\mathit{aff}}$ (both represent inward traction $-\boldsymbol{\sigma n}$). Then*

$$\|\boldsymbol{\xi}^{\mathit{full}} - \boldsymbol{\xi}^{\mathit{aff}}\|_{\widehat{\mathbf{S}}} = \min_{\boldsymbol{\mu} \in \mathbb{R}^{9m}} \|\boldsymbol{\xi}^{\mathit{full}} - \mathcal{P}_{\mathit{aff}}\, \boldsymbol{\mu}\|_{\widehat{\mathbf{S}}}, \tag{16}$$

$$\sum_i \|U_i^{\mathit{full}} - U_i^{\mathit{aff}}\|_{A_i}^2 = \|\boldsymbol{\xi}^{\mathit{full}} - \boldsymbol{\xi}^{\mathit{aff}}\|_{\widehat{\mathbf{S}}}^2. \tag{17}$$

The statements remain valid when $\widehat{\mathbf{S}}$ is only semidefinite; minimizers need not be unique but seminorm values are well defined.

Proof. Lemma A.2 reduces the Affine system to the Galerkin restriction. Subtracting $\mathcal{P}_{\text{aff}}^\top(\widehat{\mathbf{S}}\,\boldsymbol{\xi}^{\text{full}} - \widehat{\boldsymbol{F}}) = \mathbf{0}$ from $\mathbf{S}\,\boldsymbol{\lambda}^{\text{aff}} - \boldsymbol{F} = \mathbf{0}$ gives $\mathcal{P}_{\text{aff}}^\top\widehat{\mathbf{S}}(\boldsymbol{\xi}^{\text{aff}} - \boldsymbol{\xi}^{\text{full}}) = \mathbf{0}$. Hence the error $\boldsymbol{e} = \boldsymbol{\xi}^{\text{full}} - \boldsymbol{\xi}^{\text{aff}}$ is $\widehat{\mathbf{S}}$-orthogonal to $\operatorname{Range}(\mathcal{P}_{\text{aff}})$. For any $\boldsymbol{\mu}$, let $\boldsymbol{v} = \boldsymbol{\xi}^{\text{aff}} - \mathcal{P}_{\text{aff}}\boldsymbol{\mu}$. The cross term vanishes, so $\|\boldsymbol{e}+\boldsymbol{v}\|_{\widehat{\mathbf{S}}}^2 = \|\boldsymbol{e}\|_{\widehat{\mathbf{S}}}^2 + \|\boldsymbol{v}\|_{\widehat{\mathbf{S}}}^2 \ge \|\boldsymbol{e}\|_{\widehat{\mathbf{S}}}^2$. Equality at $\boldsymbol{\mu} = \boldsymbol{\lambda}^{\text{aff}}$ proves (16).

For (17), let $\delta\boldsymbol{\xi} = \boldsymbol{\xi}^{\text{full}} - \boldsymbol{\xi}^{\text{aff}}$. The local velocity difference is $\delta U_i = -H_i\big[(N_i^u)^\top \widehat{R}_i^u\,\delta\boldsymbol{\xi} + (N_i^k)^\top \delta\boldsymbol{\xi}_i^k\big]$ where $\delta\boldsymbol{\xi}_i^k$ is the boundary traction difference. Both solutions use the same prescribed boundary tractions ($\boldsymbol{\xi}_i^{k,\text{full}} = \boldsymbol{\xi}_i^{k,\text{aff}} = P_i^k R_i^k \boldsymbol{\lambda}^k$), hence $\delta\boldsymbol{\xi}_i^k = \mathbf{0}$. Only the internal-interface block survives: $\delta U_i = -H_i(N_i^u)^\top \widehat{R}_i^u\,\delta\boldsymbol{\xi}$ (the minus sign matches the load convention (5)). Using $H_i^\top A_i H_i = H_i$ from Lemma 1.2,

$$\begin{aligned}\sum_i \|\delta U_i\|_{A_i}^2 &= \sum_i (\delta U_i)^\top A_i(\delta U_i)\\ &= \sum_i \big(\widehat{R}_i^u\,\delta\boldsymbol{\xi}\big)^\top N_i^u H_i (N_i^u)^\top \big(\widehat{R}_i^u\,\delta\boldsymbol{\xi}\big) = \delta\boldsymbol{\xi}^\top \widehat{\mathbf{S}}\,\delta\boldsymbol{\xi},\end{aligned}$$

which is exactly $\|\boldsymbol{\xi}^{\text{full}} - \boldsymbol{\xi}^{\text{aff}}\|_{\widehat{\mathbf{S}}}^2$. □

Corollary A.4 (Affine versus classic enrichment). *Let $\mathcal{P}_{P0} : \mathbb{R}^m \to \mathbb{R}^{m_\Gamma}$ be the restriction to constant normal modes only (each interface contributes one column: the L^2-projection of $\varphi_0 = 1$ onto the trace basis, in the normal block-row only). Let $\boldsymbol{\lambda}^{P0}$ solve the classic Galerkin system $(\mathcal{P}_{P0}^\top \widehat{\mathbf{S}}\,\mathcal{P}_{P0})\boldsymbol{\lambda}^{P0} = \mathcal{P}_{P0}^\top \widehat{\boldsymbol{F}}$ and set $\boldsymbol{\xi}^{P0} = \mathcal{P}_{P0}\boldsymbol{\lambda}^{P0}$. Since* $\operatorname{Range}(\mathcal{P}_{P0}) \subseteq \operatorname{Range}(\mathcal{P}_{\mathit{aff}})$,

$$\|\boldsymbol{\xi}^{\mathit{full}} - \boldsymbol{\xi}^{\mathit{aff}}\|_{\widehat{\mathbf{S}}} \le \|\boldsymbol{\xi}^{\mathit{full}} - \boldsymbol{\xi}^{P0}\|_{\widehat{\mathbf{S}}}. \tag{18}$$

The same reconstruction identity (Theorem A.3, eq. (17)) applies to the classic-P0 restriction. Hence the nested-space estimate also implies monotonicity of the velocity reduction error, measured relative to the full C-DD solution, in the broken-H^1 seminorm. No analogous conclusion follows from this argument for the velocity L^2 error, pressure L^2 error, outlet-flux error, or for errors measured relative to the exact PDE solution.

A.4 Dimension summary and commutation identities

Table 1 collects the dimensions of all matrices and vectors appearing in the affine and C-DD systems. For brevity define the C-DD trace DOF counts $r_i^u := 3\sum_{\gamma\in\mathcal{U}_i} m_{i,\gamma}$ (internal) and $r_i^k := 3\sum_{\gamma\in\mathcal{K}_i} m_{i,\gamma}$ (boundary), so that $r_i^u + r_i^k$ counts all C-DD trace DOFs on pore i.

The key commutation identity linking the affine and C-DD Boolean maps is

$$\widehat{R}_i^u \mathcal{P}_{\text{aff}} = P_i^{\text{aff}} R_i^u. \tag{19}$$

Both sides map $\mathbb{R}^{9m}$ to $\mathbb{R}^{3\sum_{\gamma\in\mathcal{U}_i} m_{i,\gamma}}$. The left-hand side first applies the global affine restriction $\mathcal{P}_{\text{aff}}$ and then extracts the C-DD trace DOFs of pore i via $\widehat{R}_i^u$; the right-hand side first extracts the affine coefficients of pore i via R_i^u and then applies the local affine restriction P_i^{aff}. The equality holds because both constructions reference the same interface pairs with the same coefficient ordering.

A.5 Non-matching meshes

Assumption A.1 requires the two velocity trace spaces on each interface to coincide with a shared nodal basis. This holds when the two pore meshes share a conforming interface triangulation (the case in all numerical experiments of the companion paper, where a single global mesh is partitioned after generation).

Table 1: Dimensions of the principal objects.

Symbol	Dimension	Description
m	—	number of internal interfaces
m_i^u, m_i^k	—	internal / boundary ports of pore i
m_k	—	scalar trace DOFs on interface k
$m_{i,\gamma}$	—	scalar trace DOFs on port γ of pore i
m_Γ	$3\sum_k m_k$	total C-DD interface DOFs
$m_{\Gamma,i}^k$	$3\sum_{\gamma\in\mathcal{K}_i} m_{i,\gamma}$	C-DD boundary DOFs on pore i
$n_{u,i}$, $n_{p,i}$	—	velocity / pressure DOFs in pore i
n_i	$9m_i^u + m_i^k$	local affine DOFs in pore i
A_i	$n_{u,i} \times n_{u,i}$	velocity stiffness
B_i	$n_{p,i} \times n_{u,i}$	divergence matrix
H_i	$n_{u,i} \times n_{u,i}$	velocity–velocity block of K_i^{-1}
L_i	$n_{u,i} \times n_i$	primitive load matrix
$\mathbf{G}_i$	$n_i \times n_i$	local compliance map
R_i^u	$9m_i^u \times 9m$	Boolean map: global $\to$ local internal
R_i^k	$m_i^k \times m_i^k$	local boundary coefficient extraction (identity on pore i)
J_i^u	$n_i \times 9m_i^u$	zero extension into boundary slots
$E_{i,k}$	$9 \times 9m$	extraction of interface k's coefficient block
$N_{i,k}^\beta$	$m_k \times n_{u,i}$	C-DD coupling block, direction β
N_i^u	$r_i^u \times n_{u,i}$	C-DD coupling: all internal ports of pore i
N_i^k	$r_i^k \times n_{u,i}$	C-DD coupling: all boundary ports of pore i
$\widehat{\mathbf{S}}$	$m_\Gamma \times m_\Gamma$	C-DD Schur complement
$\widehat{R}_i^u$	$r_i^u \times m_\Gamma$	C-DD Boolean: global $\to$ local internal
P_k^{aff}	$3m_k \times 9$	affine restriction on interface k
P_i^{aff}	$r_i^u \times 9m_i^u$	block-diagonal of $\{P_{i,\gamma}^{\mathrm{aff}}\}$
P_i^k	$r_i^k \times m_i^k$	boundary affine restriction
$\mathcal{P}_{\mathrm{aff}}$	$m_\Gamma \times 9m$	global affine restriction
r_i^u	$3\sum_{\gamma\in\mathcal{U}_i} m_{i,\gamma}$	C-DD internal trace DOFs on pore i
r_i^k	$3\sum_{\gamma\in\mathcal{K}_i} m_{i,\gamma}$	C-DD boundary trace DOFs on pore i

If the two pore meshes are generated independently and do not match at the interface, the Boolean assembly maps $\widehat{R}_i^u$ must be replaced by **mortar** or **adjoint-consistent transfer** operators that preserve the symmetry and positive semidefiniteness of the coupled Schur complement. The equivalence Lemma A.2 and the Galerkin optimality Theorem A.3 would then require separate proofs under the chosen transfer scheme. The present theory does not cover non-matching interfaces.

B Statements that are *not* established

1. **Unconditional local rank.** Assumption 2.4 is not proved for general Lipschitz pores or for all meshes. Numerical observation of a single near-zero eigenvalue of $\mathbf{G}_i$ for tested geometries is evidence, not a general theorem.

2. **Per-face independence implies SPD.** This implication is false as a proof principle. The valid global result is Theorem 3.2, conditional on Assumption 2.4.

3. **Broken-pressure C-DD equals monolithic FEM.** The full-trace system uses independent local pressure spaces $Q_h^{\mathrm{br}} = \bigoplus_i Q_{i,h}$, while the monolithic Taylor–Hood problem uses globally continuous Q_h^c.

   *The "Exact FE-trace Schur" in the companion code is a primal Schur complement of the continuous-pressure monolithic system. Its agreement with the direct monolithic solve to $O(10^{-12})$ verifies the algebraic correctness of that single matrix's block elimination. It does **not** evaluate, compare, or validate the broken-pressure C-DD system used in Theorem A.3. These are two distinct discrete formulations.* Whether they produce identical velocity solutions for the tested geometries is not established or tested in the present work.

4. **Continuum or mesh-independent convergence.** All results established here are finite-dimensional and mesh-dependent. A continuum-limit analysis would require a specified h-dependent family of partitions, trace spaces, and discrete operators together with appropriate uniform stability and approximation estimates. If the interface topology varies with h (as in watershed partitions whose basin count changes with mesh size), that dependence must be incorporated explicitly into the analysis; it is not covered by the present fixed-discretisation results.

C Practical algebraic diagnostics

For each pore, the following checks directly test the hypotheses:
1. Verify symmetry and positive definiteness of A_i;
2. Verify the numerical row rank of B_i;
3. Verify $\|L_i \mathbf{1}_{i,0n} - B_i^\top \mathbf{1}_i^p\|$ at assembly tolerance (Lemma 2.3);
4. Compute singular values of $H_i L_i$ and confirm that one singular value is numerically zero (within a stated tolerance, e.g. 10^{-12} relative to the largest), with the corresponding right singular vector aligned with $\mathbf{1}_{i,0n}$ (check the angle or normalized residual, not just the singular value). As a necessary condition, verify the dimension bound $n_i - 1 \leq \mathrm{rank}(H_i) = n_{u,i} - n_{p,i}$; if this fails, Assumption 2.4 cannot hold.
5. After global assembly, verify symmetry and positive definiteness of $\mathbf{S}$ (do not infer solely from geometry);
6. Verify local mass, interface moment, and global boundary-flux residuals independently.

References

[1] S. Sun, Z. Wang, L. Zhang, J. Zhao. A domain decomposition approach to pore-network modeling of porous media flow. *arXiv:2510.13429v2*, 2026.

[2] R. A. Horn and C. R. Johnson. *Matrix Analysis*, 2nd ed. Cambridge University Press, 2012.